

# Math AA HL at KCA - Chapter 13 Notes

Tim Bao 2023-2025

February 26, 2024

## Contents

|          |                                                   |          |
|----------|---------------------------------------------------|----------|
| <b>1</b> | <b>Continuity</b>                                 | <b>3</b> |
| 1.1      | Removable and Essential Discontinuities . . . . . | 3        |
| <b>2</b> | <b>The Laws of Limit</b>                          | <b>4</b> |
| <b>3</b> | <b>Existence of Limits</b>                        | <b>4</b> |
| <b>4</b> | <b>Limits at Infinity</b>                         | <b>5</b> |

# 1 Continuity

A function is continuous if it can be drawn entirely over its domain without "lifting the pen".

The points of discontinuity are a set of points in the domain of a function at which the function is discontinuous.

## 1.1 Removable and Essential Discontinuities

There is a *removable discontinuity* at  $x = a$  if  $f(a)$  does not exist and  $\lim_{x \rightarrow a} f(x)$  exists such that a newly modified function  $g(x)$ , satisfies  $g(a) = \lim_{x \rightarrow a} f(x)$ . I.e. a new function can be defined such that the single "hole" would be "filled in" and the function is continuous at that point.

If such a modification cannot be made, then it is an *essential discontinuity*. This is when the two pieces are completely disjointed and not connected by a "hole".

## 2 The Laws of Limit

If  $\lim_{x \rightarrow a} f(x) = L$  and  $\lim_{x \rightarrow a} g(x) = M$ , then

- Additive:  $\lim_{x \rightarrow a} f(x) \pm g(x) = L \pm M$
- Multiplicative:  $\lim_{x \rightarrow a} f(x) \cdot g(x) = LM$
- Reciprocal:  $\lim_{x \rightarrow a} \left( \frac{f(x)}{g(x)} \right) = \frac{L}{M}$  if  $M \neq 0$
- L'Hôpital's:  $\lim_{x \rightarrow a} \left( \frac{f(x)}{g(x)} \right) = \lim_{x \rightarrow a} \left( \frac{f'(x)}{g'(x)} \right)$

Note that L'Hôpital's rule can be applied repeatedly.

Indeterminate forms and the corresponding ways of evaluation

- $\frac{0}{0} \longrightarrow$  factorizing, lateral limits, or L'Hôpital's rule.
- $\frac{\infty}{\infty} \longrightarrow$  comparison or L'Hôpital's rule
- $\frac{1}{0} \longrightarrow$  lateral limits

## 3 Existence of Limits

The limit  $\lim_{x \rightarrow a} f(x) = L$  exists  $\iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$ . I.e. approaching  $x = a$  from both positive and negative directions, the function converges to the *same limit*  $y = L$ .

The function diverges when there is not a limit or the limit is  $\infty$ .

## 4 Limits at Infinity

Three cases for  $\frac{f(x)}{g(x)}$

1.  $\deg(f) > \deg(g) \implies \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty$
2.  $\deg(f) = \deg(g) \implies \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$  is the quotient of the coefficients of the highest power of  $x$  of both polynomials respectively
3.  $\deg(f) < \deg(g) \implies \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$

For the second case, if both  $f(x)$  and  $g(x)$  are cubics, then, the limit is simply the coefficient of  $x^3$  in  $f(x)$  divided by the coefficient of  $x^3$  in  $g(x)$

Limit at infinity of exponentials

$$\lim_{x \rightarrow \pm\infty} e^{\mp x} = 0$$